

Enacting Molecular Interactions in VR: Preliminary relationships between visual navigation and learning outcomes

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ABSTRACT

Twenty-three undergraduates participated in a pre-post quasi-experimental single-group study involving an immersive VR simulation which allowed them to embody (i.e. become) a biomolecule and enact/experience its molecular interactions at a microscopic level using actions and gestures. Based on initial data analyses from this study, the present poster reports preliminary findings on the relationships between the participants' visual navigation, and conceptual as well as affective learning outcomes.

Keywords: Virtual reality, eye tracking, science education, embodied cognition, enactive.

1 INTRODUCTION

Head-mounted-display-based (HMD) immersive virtual reality (VR) affords novel ways of perceiving, interacting with, experiencing, and hence, learning diverse content (e.g., concepts, procedures). Specifically in STEM and allied domains, where this content is often abstract and inaccessible for direct action, the unique combination of immersion and embodied interaction that VR affords promises potentially effective pedagogies. Extant literature demonstrates the potential of VR's cognitive and affective learning effectiveness in STEM education, particularly highlighting how it helps improve comprehension, retention, problem solving abilities, transfer, and interest and engagement [1]. While newer work that suggests concrete design principles to effectively utilize immersive embodiment for STEM learning is emerging gradually [2], studies on how embodiment and enaction (e.g., of scientific phenomena) in VR, and their effects on conceptual-affective learning outcomes is limited. Similarly, work on embodiment of human-like avatars, and effects of such embodied interactions on learning are becoming more common [3]. However, work on how embodying and enacting entities and phenomena of scientific study affect learning is practically non-existent. Finally, though important to understanding VR-enabled embodied learning process, eye- and interaction-tracking investigations of STEM learners' navigation in VR environments, and relationships between visual navigation, embodiment, and learning are rare [4].

We designed an immersive VR simulation to help highschoolers and early-undergraduates learn a set of core biochemistry concepts. The VR simulation allowed learners to embody (i.e., become) a biomolecule (an enzyme) and enact/experience its molecular interactions (catalyzing biochemical reactions) at a microscopic level using actions and gestures. This poster reports preliminary findings from a pre-post quasi-experimental single group study involving twenty-three undergraduates. These findings capture change in students' conceptual understanding and attitude post simulation-play, the nature of their embodiment experience, and patterns of their eye-

gaze behavior during the simulation play.

2 THEORETICAL FRAMEWORK: EMBODIED COGNITION

Embodied cognition asserts there is a profound link between cognition, the body, and interactions with the environment. In science education, this framework suggests that active engagement with sensory perceptions and motor actions deepens students' understanding of scientific concepts, challenging traditional passive learning methods. The 4E cognition framework [5] underscores the pivotal role of contextualized physical engagement in STEM education and demonstrates how learning and understanding of even abstract content is built on top of one's physical interactions with content [6, 7].

Newer immersive technologies such as VR allow students to immerse themselves in scientific subjects and allow realistic bodily engagement (e.g. manipulating virtual objects and conducting experiments; [2]). Such engagements transform learning into a multisensory encounter, and thus generating the potential enhance an intervention's overall learning effectiveness, in both cognitive and affective terms [8].

3 METHODS

Twenty-three students enrolled in life sciences programs at a major Danish university volunteered to participate in a one-group pre-post study.

We used an HTC Vive Cosmos HMD (tethered to a DELL gaming laptop) to deploy the simulation. A VR eye-tracker add-on (Pupil Labs) was installed inside the VR HMD to capture students' gaze-data.

Each student took a pre-test on their [prior understanding of the biochemistry content](#) presented in the simulation, and [intrinsic motivation and self-efficacy](#) in relation to learning biochemistry. The participant then wore the HMD, followed on-screen instructions for eye-tracker calibration, and then began interacting with our embodied VR biochemistry simulation (~15-25 minutes). Finally, each student responded to a post-test comprising of the same items as the pre-test, plus a self-report 18-item questionnaire (5-point Likert scale) on their embodiment experience.

The study was conducted in accordance with the Declaration of Helsinki and the general ethical guidelines of the university. All participating students were adults and were informed about the study's objectives prior to recruitment. Written consents were obtained from the participants before the study.

3.1 Instructional Materials

Our VR simulation narrates a classic case of subcellular biochemical interactions (enzyme-substrate reactions). Various visual prompts, text-based instructions, and dynamic haptic feedback embedded in our simulation allows a learner to experience (a) becoming an enzyme molecule (visible as molecular arms whose movement is synced with the learner's), (b) navigate bacterial cell wall structures at molecular level, and (c) enact biochemical reaction mechanisms through distinct controller actions/gestures (see [9] for more details).

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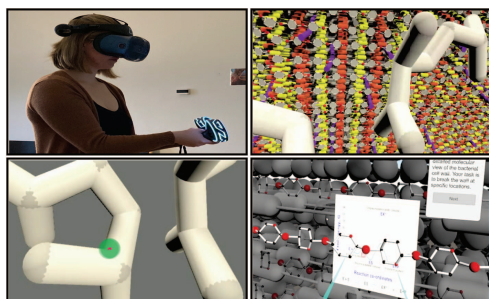


Figure 1: Participant using the headset and controllers (top-left). Embodied enzyme arms (top-right & bottom-left). Participant interacting with molecules at the molecular level (bottom-right).

3.2 Analysis

Due to partial data for one participant, our effective sample size for data analysis involved 22 participants.

The following means were calculated from participant responses to the pre- and post-tests and ratings on post-intervention questionnaire to check extent of embodied experience: (i) knowledge gain score, (ii) intrinsic motivation and self-efficacy gain scores, and (iii) embodiment score. Wilcoxon signed-rank test was performed on pre-post test scores.

The eye-tracker yielded gaze data for each participant; these data were collected and exported for further analyses using Pupil Player [10]. For each participant we had video data of their navigation in the VR simulation on which their fixation data was overlaid. From this, we calculated fixation metrics for each participant, such as total number of fixations (TNF), total fixation duration (TFD), and normalized fixation counts per second (TNF/TFD). The simulation had 3 chronological stages - Before Embodiment of the enzyme, After Embodiment, and Tasks. We calculated TNF/TFD (i.e. normalized fixation counts) for each stage. A correlation analysis was conducted in R Studio (*psych* and *corr* packages) to examine preliminary relationship between TNF/TFD parameter of visual navigation per stage and gain scores.

4 FINDINGS

Preliminary testing showed that: (i) the students' conceptual understanding increased ($p < 0.0001$) after the VR experience (pre-test: $M = 4.09$; $SD = 1.56$; post-test: $M = 6.78$; $SD = 1.86$), (ii) there were no statistically significant changes in students' motivation (pre-test: $M = 3.39$; $SD = 1.06$; post-test: $M = 3.84$; $SD = 0.89$) and self-efficacy (pre-test: $M = 2.14$; $SD = 0.89$; post-test: $M = 3.11$; $SD = 1.17$). Finally, (iii) our simulation was moderately effective in helping students bodily experience embodiment of biomolecule at the molecular level (Mean = 2.9, $SD = 0.58$; on a 5-point Likert scale).

Correlation Matrix	Knowledge gain score	Motivation gain score	Self-efficacy gain scores	Total embodiment score
Before Embodiment	0.46*	-0.07	-0.34	-0.06
After Embodiment	0.36	-0.47*	-0.55*	-0.23
Task	0.43*	-0.34	-0.55*	0.00

Table 1: Correlation matrix of the VR simulation stages and the performance outcome measures.

For eye-tracking, the average participant total fixation count (TFC) was 2,832.64 and the average participant total fixation duration (TFD) time was about 402.26 seconds (or 6.7 minutes).

After running a Pearson correlation, we found the following, fixation count/sec is correlated with learning gains but negatively correlated with affective outcomes (see Table 1; measures with significance of $p < 0.05$ are indicated with an asterisk, all other correlation values did not have a significant p-values).

5 DISCUSSION AND CONCLUSION

This study utilized an HTC Vive Cosmos HMD and VR eye-tracker to investigate the impact of engaging twenty-three life sciences students in a unique embodied and enactive VR experience of molecular interactions on their learning of core biochemistry concepts. Preliminary analyses indicated that enactment in VR helped improve students' biochemistry understanding significantly, thus, highlighting VR's effectiveness in promoting embodied learning experiences. Further, the correlation analyses revealed nuanced relationships between gaze behavior at different stages of embodied interaction in VR and the learning outcomes. This research addresses gaps in the embodiment of scientific concepts, contributes valuable insights into STEM education and VR technologies.

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